Biased Algorithmic Decision-Making

Leadership, Explainability, and Harm in AI-Powered Hiring Models

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CODE & REPRODUCIBILITY

Scan for the GitHub repository with the full analysis workflow, notebooks, model audit code, SHAP / LIME explainability, visuals, & poster figures.

UPSTREAM:
WHERE GENDERED HARM BEGINS

Job Advertisements
Masculine-coded wording is a deterrent, especially in leadership roles.

Historical Data
Past hiring records encode and amplify structural gender inequalities.

Workforce & Society
Unequal opportunities shape the data that trains the model.

AI HIRING PIPELINE



MODEL DESIGN AND TRAINING CHOICES

Features | Training Data | Evaluation | Thresholds | Deployment

EXPLAINABILITY & AUDITING



IMPACTS

Filtered Out Candidates
Who is overlooked?

Selected Candidates
Who gets in?

Gendered Harm
Lost talent, widened inequality, weaker organizations.

Banner graphics generated with ChatGPT (OpenAI, 2026).

1. ABSTRACT

AI-powered decision-making systems promise efficiency and scalability, yet they can unwittingly reproduce and amplify historical gender inequalities. This poster examines biased algorithmic decision-making as a socio-technical leadership challenge.

Bias can enter the machine learning pipeline upstream through job advertisements, historical patterns, and proxy variables in the data, and persist through model design, feature evaluation choices, and deployment (Wange et al., 2019). The Amazon Hiring Case illustrates how models trained on historical data systematically penalized female-coded signals and were abandoned when mitigation attempts failed (Chang, 2023; Rodriguez, 2023).

Balancing the data alone is not a sufficient mitigation tactic, as proxy variables and data leakage can preserve biased patterns even when a protected attribute is removed (Fabeyo, 2025). The explainability tools, such as Shapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), improve transparency into the algorithmic decision-making process post-hoc. However, explainability is not the same as fairness.

Effective mitigation of this issue requires interdisciplinary leadership, inclusive language practices, diverse teams, fairness thresholds, auditing, explainability reviews, and human oversight (Schriesheim et al., 2006; Mekonnen & Bayissa, 2023).

Ultimately, biased algorithmic decision-making is not only a technical challenge but a leadership responsibility, requiring leaders to set ethical priorities, build accountable systems, and ensure that AI adoption advances equity rather than reproducing historical harm.

2. RESEARCH FOCUS

How does bias travel through artificial intelligence/machine learning pipelines, with a focus on AI hiring pipelines, and how should leadership disrupt the harm it produces?

3. LITERATURE & CASE CONTEXT

AI systems learn from historical data shaped by structural inequality, meaning models can reproduce the cultural standards and inequities already present in the data (hooks, 2015; Chen, 2023).

Pre-model bias in job advertisements can discourage women from applying, particularly when masculine-coded wording appears in leadership or senior-role descriptions (Ridgway et al., 2022; Gaucher et al., 2011; Hu et al., 2022).

Proxy variables and data leakage can preserve gendered opportunity structures even when protected attributes are removed from the model (Wang et al., 2019).

Amazon's recruiting tool learned to penalize women-coded signals from resumes and was abandoned after mitigation efforts failed (Chang, 2023; Rodriguez, 2023).

Leadership research shows that technical mitigation alone is insufficient; equitable AI adoption requires transparency, governance, accountability, and leadership practices that pair transformational vision with transactional guardrails (Fabeyo, 2025; Schriesheim et al., 2006; Mekonnen & Bayissa, 2023).

4. METHODS: SHAP & LIME EXPLAINABILITY DEMO

GOAL

Demonstrate how predictive systems infer and reproduce biased labor-market outcomes.

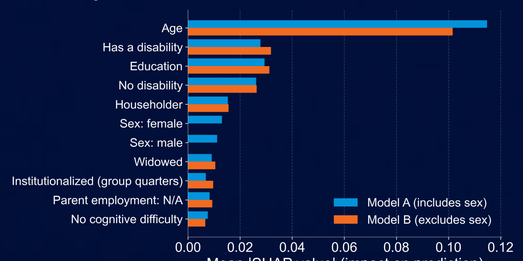
POPULATION

2024 American Community Survey (ACS) Public Use Microdata Sample, California, accessed via folktables. N = 328,738 people aged 16–90 (57.1% employed). Target: employed vs. not employed, a labor-market proxy rather than actual hiring decisions.

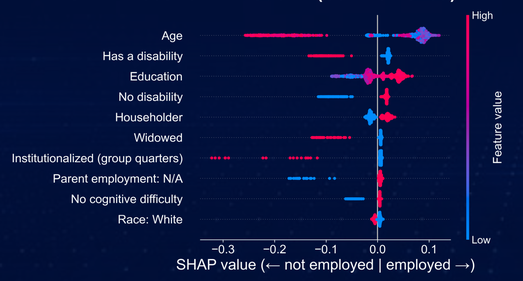
MODELING

- Stratified sample of 59,999 people; 80/20 split (47,999 train / 12,000 test).
- Random forest (150 trees). Model A includes sex and race (accuracy 77.6%). Model B removes sex but keeps it for auditing (accuracy 77.4%).
- Audits: four-fifths (0.80) rule by sex and by sex × race, plus a counterfactual sex-swap test.
- SHAP (global) and LIME (local) explain how the models make predictions.

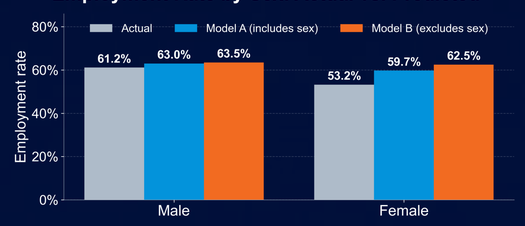
Top SHAP Features: With vs. Without Sex



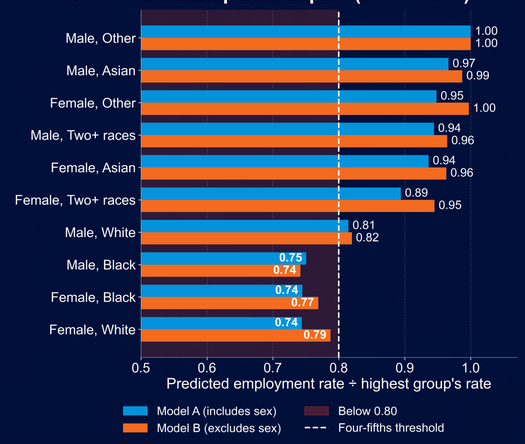
SHAP Beeswarm: Model B (Sex Removed)



Employment Rate by Sex: Actual vs. Predicted

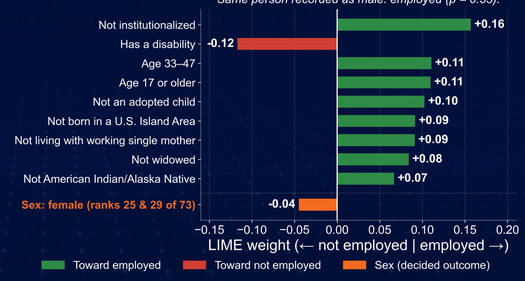


Intersectional Disparate Impact (Sex × Race)



LIME: Counterfactual Test

Employed woman, 35: predicted not employed (p = 0.49). Same person recorded as male: employed (p = 0.53).



5. KEY FINDINGS

Employment is already uneven by sex: 61.1% of men vs. 53.3% of women are employed.

By sex alone, both models pass the four-fifths rule (female/male ratio: 0.95 with sex, 0.98 without).

By sex × race, Model A fails for White women (0.74), Black women (0.74), and Black men (0.75).

Removing sex does not fix it: in Model B, the same three groups stay below 0.80 (0.79, 0.77, 0.74).

Removing sex barely changes accuracy (77.6% vs. 77.4%).

Counterfactual test: recording men as female lowered predicted employment 4.6 points on average and changed 2.5% of decisions; for 87 employed women, sex alone meant "not employed."

6. LIMITATIONS

1. The ACS serves only as a labor-market proxy, not actual hiring data, so these results show associations and cannot prove causality or discrimination.
2. This analysis focuses on hiring as one domain of biased algorithmic decision making; similar harms may appear differently across education, finance, healthcare, or policing.
3. The poster identifies leadership responsibilities but does not test a specific leadership intervention within an organization.
4. Organizational incentives, legal requirements, vendor constraints, and HR practices may limit how easily fairness recommendations can be implemented.
5. Explainability tools can support accountability, but their usefulness depends on whether leaders and decision-makers are willing to act on what the tools reveal.
6. Intersectional subgroups are small (Black women and men, n ≈ 256 each; about ±6 points), results come from one California sample and one train/test split, and sex is recorded as binary.

7. LEADERSHIP IMPLICATIONS

Bias mitigation is a leadership responsibility, not only a technical task. Leaders decide what counts as fair, who checks it, and what happens when a system falls short, pairing transformational vision with transactional guardrails.

1. Ask "fair for whom?" Audit across intersecting groups (e.g., sex × race), not one attribute at a time.
2. Removing protected attributes is not a fix: dropping sex barely changed accuracy and did not close the gaps. Require proxy reviews.
3. Own the standard: leaders, not vendors, set fairness thresholds, audit schedules, and escalation steps before deployment.
4. Treat explanations as prompts for questions, not proof; LIME ranked the factor that decided one woman's outcome 25th.
5. Lead for scale: small gaps affect thousands of applicants, so monitor outcomes continuously and keep a human in the loop.
6. Start upstream and together: audit job ads and training data with diverse teams across HR, data science, ethics, and legal.

8. CONCLUSION

Biased algorithmic decision-making in hiring is a systemic sociotechnical problem. Fairness checks that examine one attribute at a time can miss disparities that emerge at identity intersections, even when protected attributes are removed. Equitable hiring requires intersectional auditing, human oversight, and accountable leadership.

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